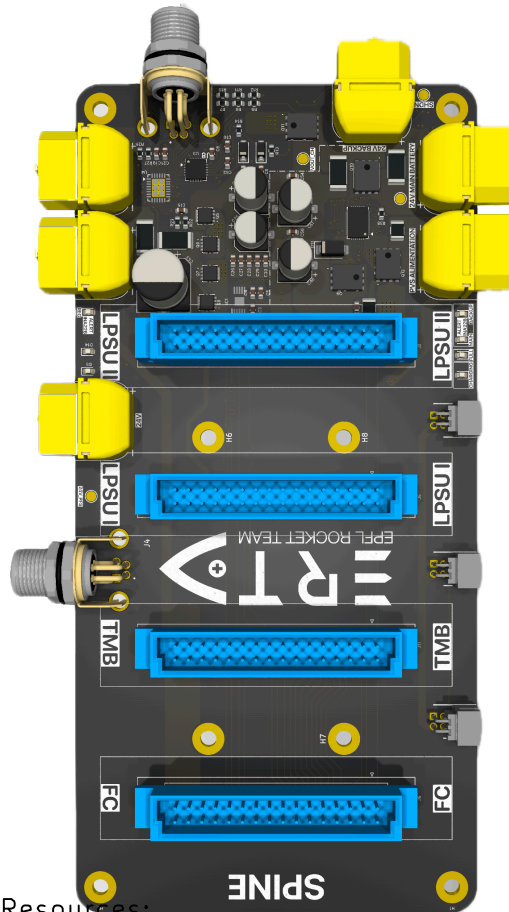
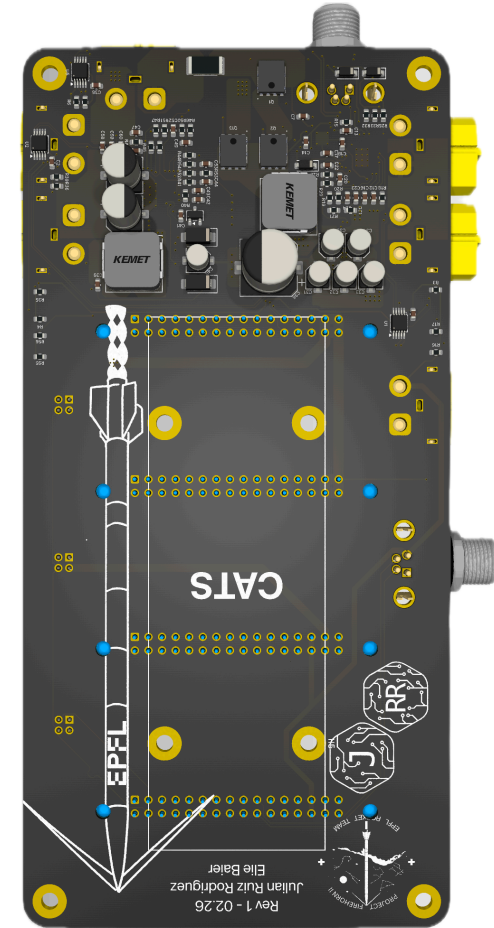


# FH2 – Spine

TOP VIEW



BOTTOM VIEW



## External Resources:

- Power Sizing Excel
- DIN41612 Connectors Layout

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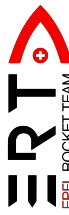
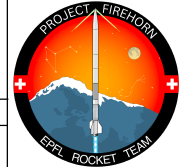
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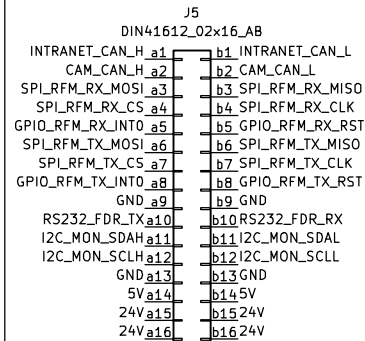
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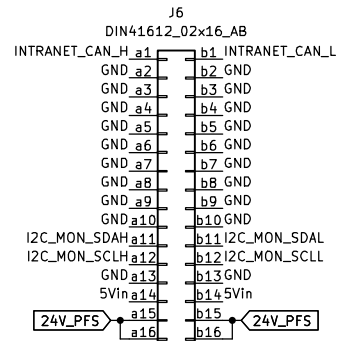
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|---------------------------------------------------------------------------------------|---------------------------|---|-----------|---------------------------------------------------------------------------------------|----------|
|  | <b>EPFL Rocket Team</b>   |   |           |  |          |
|                                                                                       | Author: Elie Baier        |   | Reviewer: |                                                                                       |          |
|                                                                                       | Sheet: /                  |   |           |                                                                                       |          |
|                                                                                       | File: spine.kicad_sch     |   |           |                                                                                       |          |
|                                                                                       | <b>Title: FH2 – Spine</b> |   |           |                                                                                       |          |
|                                                                                       | KiCad E.D.A. 9.0.7        |   |           |                                                                                       | Size: A4 |
| 4                                                                                     |                           | 5 |           | Id: 1/6                                                                               |          |

# Page Title

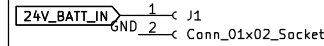
DIN41612 FC Connector



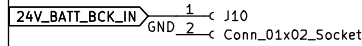
DIN41612 LPSU Connector



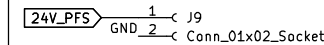
XT60 Power IN Battery



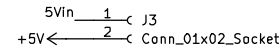
XT60 Power IN Battery Backup



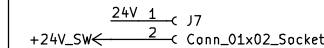
XT60 Power IN PFS



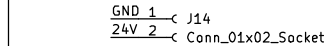
XT60 Switch LP



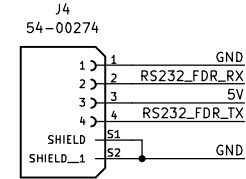
XT60 Switch HP



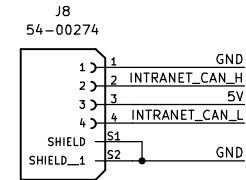
XT60 24V OUT



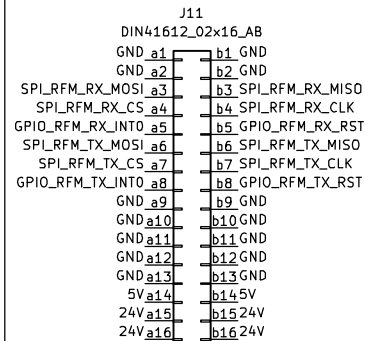
SS-12800 Circular Connector 1



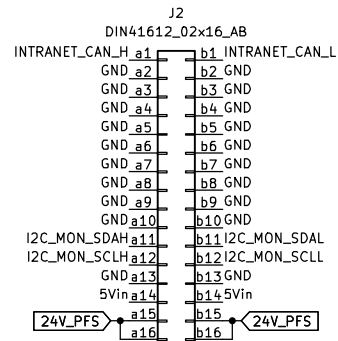
SS-12800 Circular Connector 2



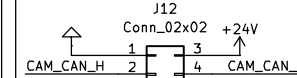
DIN41612 Telemetry Connector



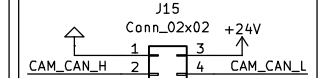
DIN41612 LPSU Connector 2



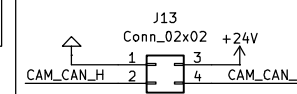
Cam Connector 1



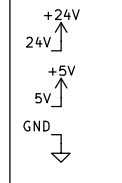
Cam Connector 3



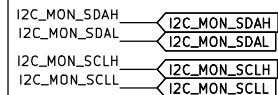
Cam Connector 2



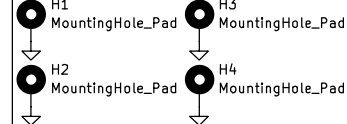
Global Power



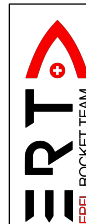
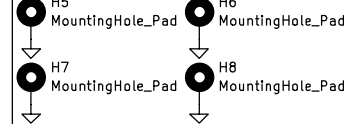
Global I2C



SPINE



CATS



## EPFL Rocket Team

Author: Elie Baier

Reviewer:

Sheet: /Connectors/

File: connectors.kicad\_sch

Title: Connectors

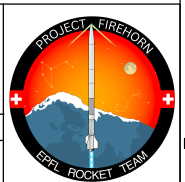
KiCad E.D.A. 9.0.7

Size: A4

Date: 2026-03-01

Rev: v1

Id: 2/6

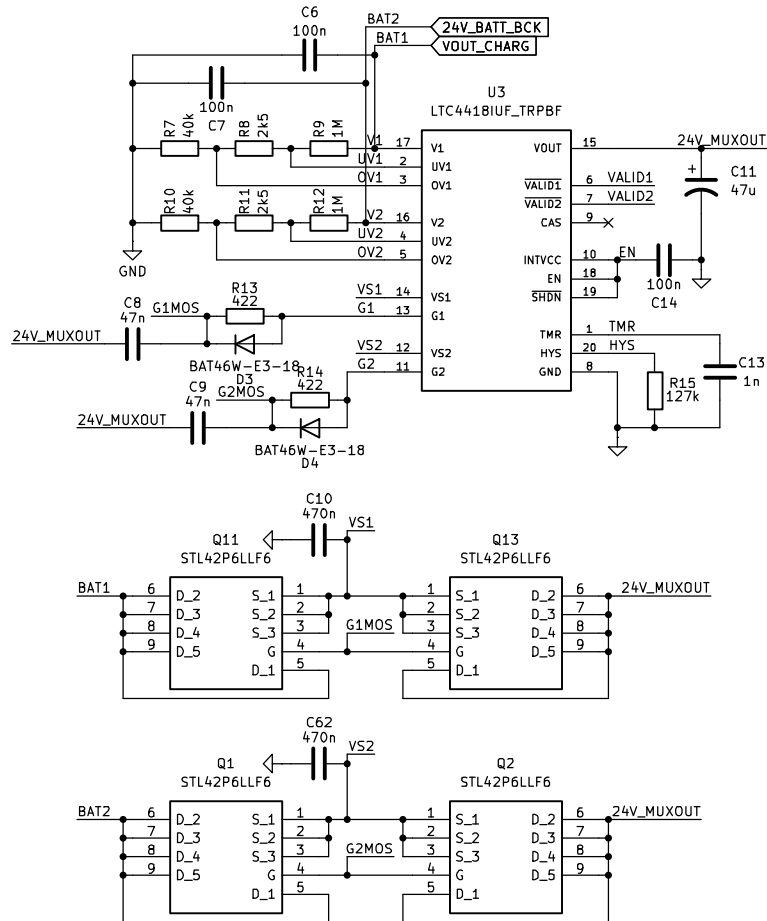


# Page Title

|                                                                                       |                           |          |                  |                                                                                       |
|---------------------------------------------------------------------------------------|---------------------------|----------|------------------|---------------------------------------------------------------------------------------|
|  | <b>EPFL Rocket Team</b>   |          |                  |  |
|                                                                                       | Author: Elie Baier        |          | Reviewer:        |                                                                                       |
|                                                                                       | Sheet: /Power/            |          |                  |                                                                                       |
|                                                                                       | File: power.kicad_sch     |          |                  |                                                                                       |
|                                                                                       | <b>Title: FH2 – Spine</b> |          |                  |                                                                                       |
| KiCad E.D.A. 9.0.7                                                                    |                           | Size: A4 | Date: 2026-03-01 | Rev: v1                                                                               |
|                                                                                       |                           |          |                  | Id: 3/6                                                                               |

# Page Title

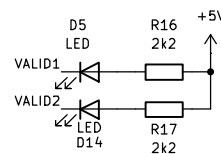
## Power MUX



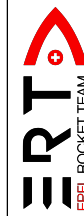
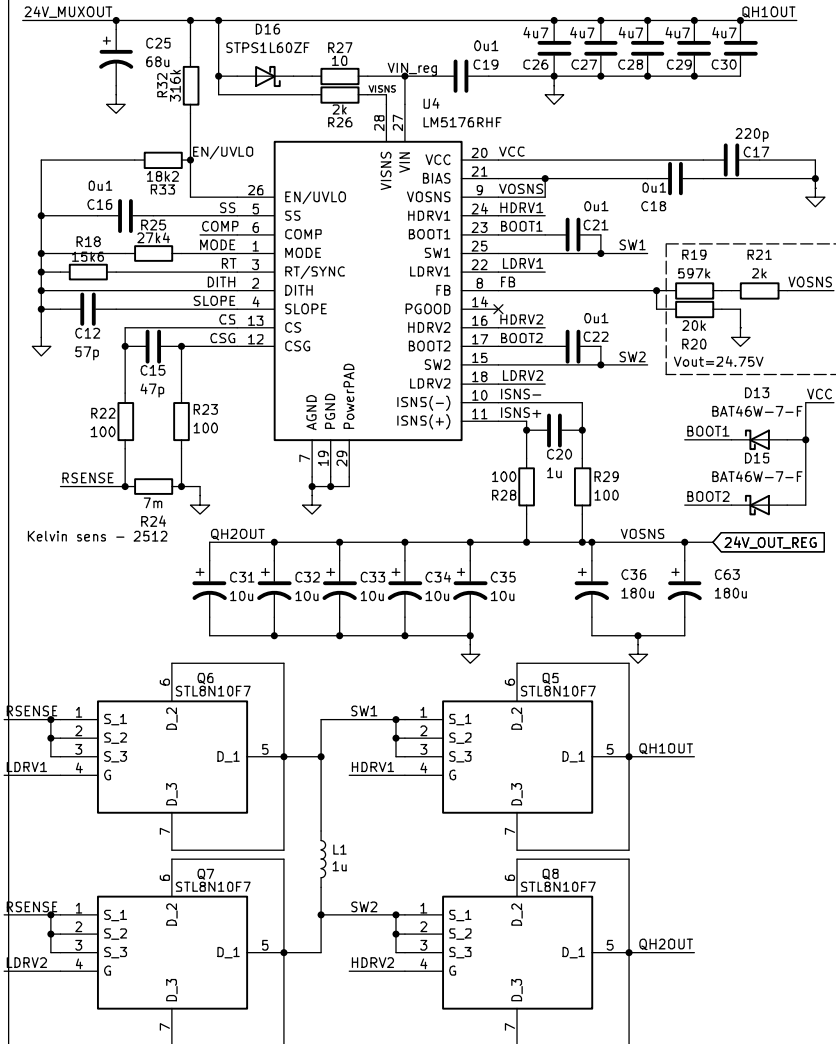
- Notes:
- V1 always higher priority
  - Stable for 16ms for a source to be valid
  - 0.5V hysteresis
  - PMOS should be at 40 V VDS and RDS(on) at VGS ≈ -4.5 to -6 V (IPB80P04P4-05)

## Source LEDs

- Notes:
- High when source detected as valid



## 24V Regulator



## EPFL Rocket Team

Author: Elie Baier

Reviewer:

Sheet: /Power/Management/  
File: power\_mgmt.kicad\_sch

Title: FH2 - Spine

KiCad E.D.A. 9.0.7

Size: A4

Date: 2026-03-01

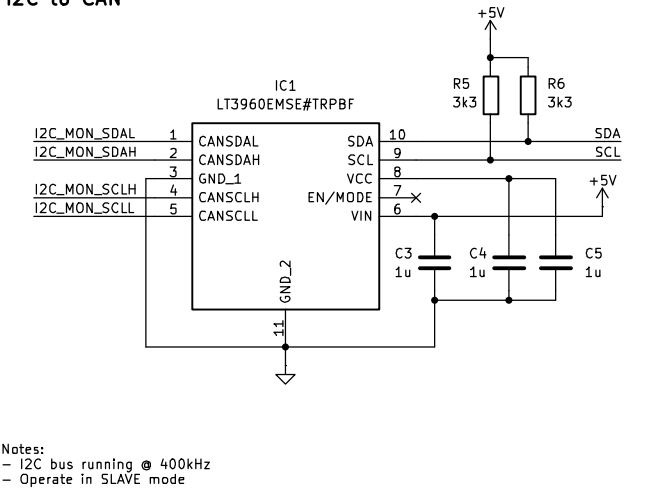
Rev: v1

Id: 4/6



Page Title

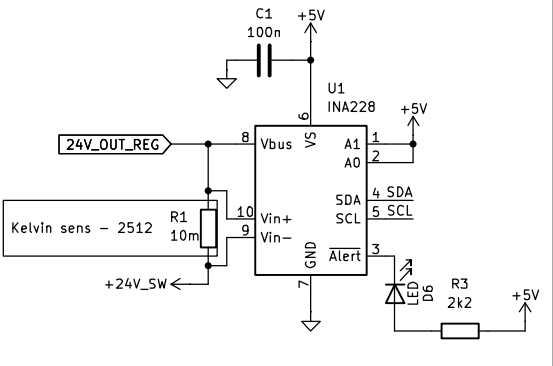
I2C to CAN



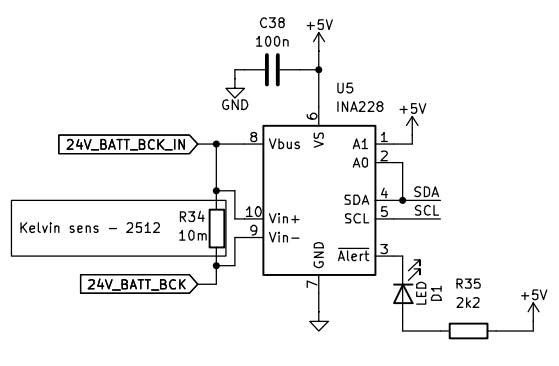
Notes:  
- I2C bus running @ 400kHz  
- Operate in SLAVE mode

|              |              |
|--------------|--------------|
| Global I2C   |              |
| I2C_MON_SDAH | I2C_MON_SDAH |
| I2C_MON_SDAH | I2C_MON_SDAH |
| I2C_MON_SCLH | I2C_MON_SCLH |
| I2C_MON_SCLH | I2C_MON_SCLH |
| I2C_MON_SCLL | I2C_MON_SCLL |
| I2C_MON_SCLL | I2C_MON_SCLL |

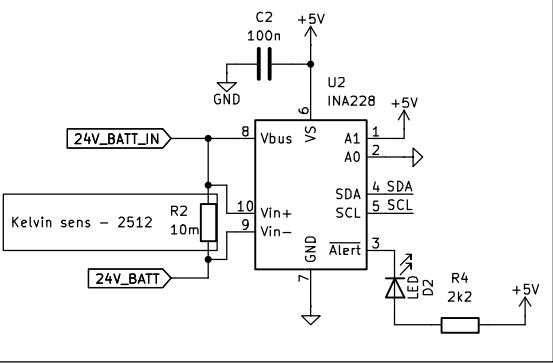
REGU. OUT MONITORING



BAT2 MONITORING



BAT1 MONITORING



EPFL Rocket Team

Author: Elie Baier

Reviewer:

Sheet: /Power/Monitoring/

File: power\_mon.kicad\_sch

Title: FH2 - Spine

KiCad E.D.A. 9.0.7

Size: A4

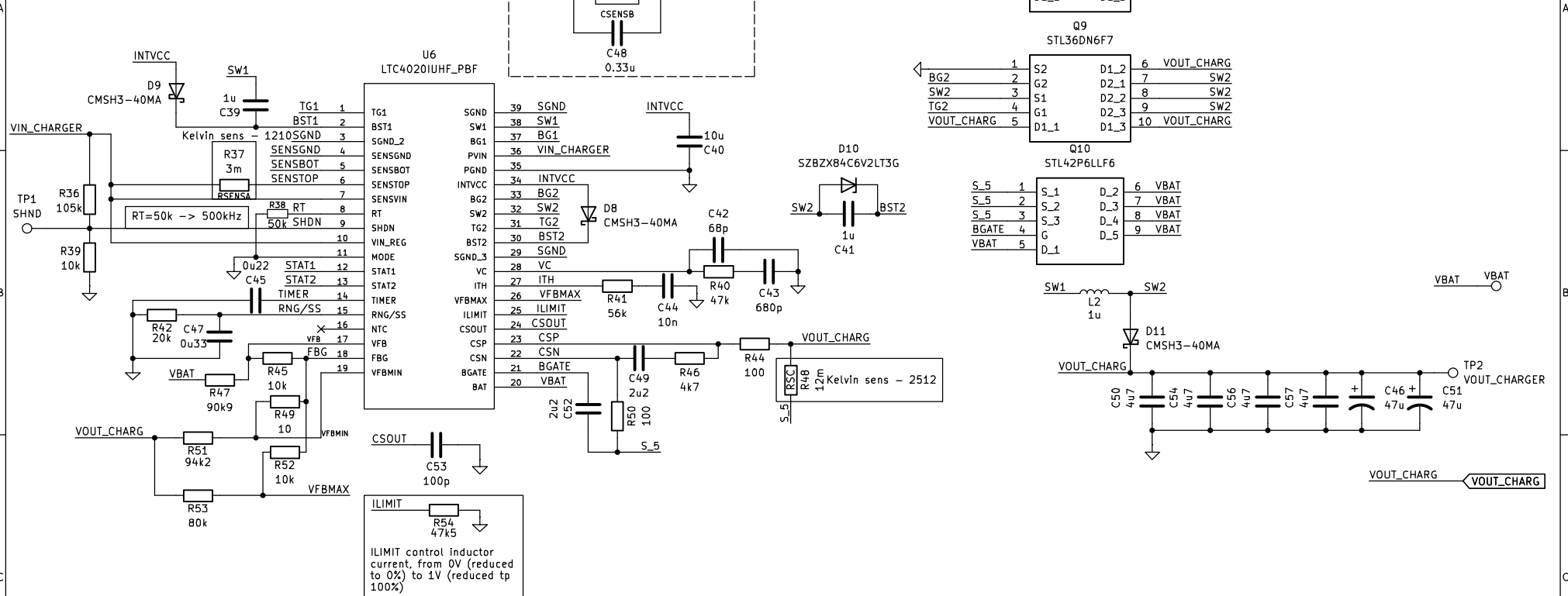
Date: 2026-03-01

Rev: v1

Id: 5/6

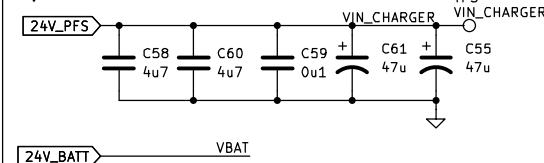


# Page Title

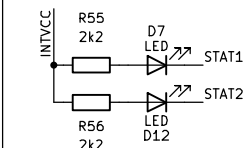


- Notes:
- BAT. Vfloat (RFB) is given by  $2.5 \cdot (1 + (R47/R45)) \Rightarrow 2.5 \cdot (1 + (90.9/10)) = 25.225$  (vs. 25.20 in 2026\_C\_SE\_PTS)
  - VOUT min. (RMIN) given by  $2.125 \cdot (1 + (R51/R49)) \Rightarrow 2.125 \cdot (1 + (94k2/10)) = 22.14V$
  - VOUT max. (RMAX) given by  $2.75 \cdot (1 + (R53/R52)) \Rightarrow 2.75 \cdot (1 + (80k/10k)) = 24.75V$
  - BAT. max. I (Input) set to 5A max.  $\Rightarrow 0.05/R48 = 0.05/12m = 4A$
  - VIN max. I set to 10A max. (5A BAT. + 5A load)  $\Rightarrow 0.05/(R3 = R10) = 0.05/0.005 = 10A$
  - TIMER set to 3h to charge for safety
  - NTC (BAT. Temp. Monitoring) disabled (floating) (D5 p.10)
  - C22 gives a current ramp in 6-7ms

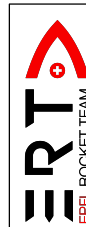
## Inputs



## STATUS LED



- Notes:
- D1 active if charging in progress
  - D2 active if charging complete or fault



## EPFL Rocket Team

Author: Elie Baier

Reviewer:

Sheet: /Power/Charger/

File: power\_charger.kicad\_sch

Title: FH2 - Spine

KiCad E.D.A. 9.0.7

Size: A4

Date: 2026-03-01

Rev: v1

Id: 6/6

